

ActInf GuestStream #035.1 ~ Jordan Hall & Matthew Pirkowski

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foreign hello and welcome this is active inference guest stream number 35.1 it's January 24th 2023. today we're here with Jordan Hall and Matthew burkowski we're going to have a very interesting discussion so thank you both for joining we will begin with a Hello introduction from Jordan Matthew will then set some context that will launch us into a dialogue that will take us on through so thanks again and Jordan please um since I lied in the Preamble so a little bit of introduction I'm noticing and this is I think part of the real interest of having a conversation with Matthew um uh is something like it would be nice I think and plausible for us to actually produce something new and useful in this 90-minute time span and I expect that we'll want to be using something like a a transparadigmatic method so we should be grabbing Concepts from a variety of different uh disciplines and frames and applying them in the area of inquiry and endeavoring to use that to actually identify something that is specific and perhaps hasn't been as articulated or deeply investigated at least to my awareness of course um before and so the end result hopefully will actually be worth worth actually having been done and to have been participated in that's probably I I mean I I hope that at least this recording in and of itself as an artifact will be something useful for a number of people who uh who might have been paying attention to the initial context and that we can bring together as we typically do uh the many types of frames that we've been exploring uh in between our conversations that are that are quite uh quite fertile and oftentimes barefooter in conversation so I'm always I always look forward to our conversation specifically for that reason why we are here this morning or I guess what did catalyze this initially was a tweet that kind of randomly sent out the other day as I had a network of Concepts floating around in my head around uh emergence governance the Tendencies uh toward which that pattern flows and so the Tweet may be the best to just read it um so because that all forms of governance Trend toward parasitic corruption should be the first quote-unquote law of political quote-unquote science the only questions are how long before this becomes unsustainable does it also generate valuable

byproducts and will it will it fail catastrophically or adaptively and so you know just the general frame for the conversation is something along the lines of as we move through the world together and attempt to identify points of tension uh in terms of the relationship between our communities and the embedding context in which they exist and attempt to navigate the world such that they can retain their ability to to function to cohere um to generate new solutions that help them mediate the tensions in the world that are constantly attempting to or constantly acting as a force against those structures as we come together to try to to to um to network ourselves such that we can function adaptively right to create systems for doing so and that's evolved you know quite a bit over time from uh informal systems at the tribal level at smaller social scales uh with respect to uh structures that had much greater coherence just as a function of the fact that everyone knew each other everyone was around one another uh in a daily context uh had a deep history in terms of their relational contextual knowledge of uh those interpersonal relations these lineages and so you didn't need a great deal of abstraction in the quote-unquote political domain you didn't need to conceptualize an abstract structure and then attempt to convince people that this was the legitimate structure to buy into as a stewardship mechanism for themselves but as we scaled up you know we started to introduce these sorts of abstractions um and and we are certainly in a time now where you know it's worth asking this question is as we create abstractions that are uh into which we pour our political energies you know for example the idea of democratic governance or you know the the sort of Republic that uh we live in in the United States to what extent is their tendency uh of those who are given stewardship over that given their placement given their necessity necessary placement within this mechanism but also uh animate stuff they have a mechanism but also just the structure of that mechanism uh being responsible as a small subset of the population for stewarding its large abstraction uh and this large degree of attention and this large degree of responsibility uh is it possible for that to be resistant to corruption across long periods of time is it inevitable that uh that corruption will exist and this was the claim in the Tweet uh that people will fundamentally begin to make choices that uh that that direct all of that attention energy toward their own ends as opposed to the ends ostensible ends of the collective um and then if that is a given what frames can we hold despite the fact that this is a given that can still act as evaluative criteria and so a useful evaluated criteria which is what sort of those three ideas um that I provided related to in terms of the degree to which that can be an unsustainable or a sustainable structure um to the extent that it still generates valuable byproducts even though that that corruption is necessary or inherently emergent within it or

um whether it will fail catastrophically or adaptively will it essentially uh will it undermine itself entirely or will it actually within itself despite that corruption uh be able to surface new patterns into which itself can fail uh as a kind of Lifeboat that increasingly expands its own capacity so that was you know I'll kind of step away now and just give time for a reflection and um analysis on y'all's part um but that was just sort of the initial frame that I think catalyzed some conversation here all right well there's a there's a whole bunch of different things that um are I think implicated and I noticed that I want to just kind of put a put a variety of different uh plates on the table without having to understand exactly what meal we're eating just yet so in no particular order one thing that comes up is I feel like there's a there's an invocation of a frame that I've used a lot and that is something along the lines of like major eras or apocs in the Arc of evolution specifically in the context of humans and so the first era um wide which I which I refer to often as unconscious Evolution or ordinary Evolution what is the is the period I think you were referring to be something like lineage where there's the The Binding constraints the various kinds of characteristics of uh behavior that steer the collective Choice making of a group of sort of indigenous humans uh was largely a consequence of a very long time and iterations under the ordinary constraints of what I'll just call ordinary biological evolution so lots of trial and error and conservation of relatively fit behaviors very tightly knit and with all the various advantages that are associated with that um modality of exploring possibility space the second year of the second apoc would be conscious design and I think this is the domain where we see most of the problematics and I'll invoke people like tainter and [] **as other individuals who've pointed to the problematics that seem to be intrinsic to the apoc or the modality of conscious design and of course one of the sort of most fundamental challenges is that it exits us from the equilibrium or the homeostatic feedback loops that are characteristic of unconscious Evolution so humans are able to operate in the mode of conscious design we begin to create novelty in the environment that produces asymmetry in relationship with all the other feedback loops that we happen to have been embedded in contextually but we don't actually fully understand the consequences of the asymmetries that we're producing and this creates a whole series of cascade effect so change explores that in the context of how a designed social infrastructure produces an arc that has a built-in closure because as we sort of play out the entropy that was built into the Inception of that social infrastructure which may take Generations or even many many generations we find ourselves collapsing back into the complex domain of the unconscious evolutionary basis and then we have a reset I think**

[] more or less explores the same basic Dynamic within the particular domain of Science and I think they're they're largely if you look at them as being examples of a more generalized problematic you can then or or even Dynamic you can then use that to create a meta Theory similarly I'm not sure exactly who studied this most carefully perhaps Hayek looking at it in the context of businesses which have a very similar and very well studied Dynamic the the s-curve of a business structure where an entrepreneurial um group usually one or two or five Founders like a very small group operating in many ways principally from the unconscious evolutionary stage like operating in very much a human indigenous modality using a lot of the basic human dynamics to be in deep relationship with each other and form a very fluid open coherent governance um highly generative produces a large gradient or a large well of potential but then as it structures itself into the conscious design mode it starts creating an organizational structure it plays the various tools and in many ways kind of replicates The Arc of social structures in general recreating the rise and fall of civilizations and then form some form of closure that eventually dies since I think it's studied pretty thoroughly in a number of different locations and given that languaging and I haven't actually brought in parasite which is a very good you know so it's just bringing parasite and the notion of Niche right so a general principle of evolution is that every Niche will be ultimately explored and there's something about both the presence of entropy in design constructs or actually any any kind of uh gradient or any any kind of gradient has a dynamic of entropy but it also has the dynamic of once the gradient is produced that gradient represents a niche and the exploration and exploitation of that niche of which parasitism is one uh fundamental strategy seems to be something that needs to be we need to be aware of and think about how that's addressed across different apocs um and then the the last piece I would throw out is something like okay uh if we're endeavoring to move into a third mode which would be something like the apoc of conscious evolutionary approach which is able to sort of combine the best of the both and perhaps avoid the negatives of both by the way entirely by hypothesis they have no reason to believe this is I have no compelling reason to believe this is actually plausible one of the things that might be interesting is to look at the phenotype genotype Dynamic I.E Evolution seems to have woven at a very deep level uh what I'll call a modal a very profound modal shift which is that all phenotypes die but there's continuity in the phenotype produces uh a embodied version of the genotype so the genotype has continuity exclusively however mediated by the phenotype and the phenotype has a kind of continuity but exclusively mediated by the genotype and the two are very very distinct in terms of their physical characteristics right the things that are

associated with energetic gradients are qualitatively distinct and the shifting back and forth between the two seems to be a solution to the kind of problem that we're actually talking about um okay so that's what I'm putting into the mix into the stew number of ingredients to work with um yeah so beginning at the beginning of your statement with respect to uh the introduction of [] uh hike and um tainter so the just give me some grammatic work I think we have to do as well here because there's uh he used the word closure a couple of times and uh the manner in which he used it uh was sort of uh and sort of the epoch ending book and context uh in a way is that does that is that a fair characterization of how you you felt you were using it in a way is that sort of yeah it has like the closing of uh the closing of an era the end of an error the end of a phase I think even even more pointedly to the notion of a uh a thermodynamically closed system these kinds of systems the proposition I'm making is that they are closed systems and so therefore they have a uh a finite time until they achieve something like an entropic equilibrium and that sort of does that that Arc of the playing out of that time is the story of their of their of their particular I would propose okay interesting yeah and I mean so there's always there's always there's this fascinating tension along the lines of sort of uh stable or unstable equilibriums or you know if you if you want to layer in another I'm not sure how familiar you are the work of Terence Deacon but he he's quite interesting in so far as he provides thermodynamic Frameworks around the ideas of uh different um emergent behaviors that flow from sort of this first homeodynamic level which is systems that tend to be uh um that tend to essentially equilibrium so so he introduced a little bit of language called the ortho grade as a sort of abstract general term of like a number of systems that tend toward a uh a more homogeneous State let's say or a more stable State and then Contra grade which is essentially the capacity of any system to given an ambient orthograde context to to sort of act against that in some way um continuously or with some degree of stability and he builds a framework of um you know actually showing that certain types of um structure itself and this kind of resonates with a lot of Kaufman's work in terms of uh the the origins of order looking for non-evolutionary just uh physical constraints that give Horizon processes that interact with those physical constraints that themselves give rise to certain forms of order in the world so uh you know Deacon's claim is something along the lines of two types of systems that might themselves tend toward stable more dissipative structure if put in relationship in a particular way with one another can then give rise to at that sort of frustration boundary

afloat through that frustration boundary give rise to some sort of actual structure in the world or pattern in the world and the architectural example he uses there is something called like a Rayleigh Bernard cell which is heat trying to our energy trying to flow through uh water or oil at a with a particular viscosity um and you know the first thing you get is the actual bubbling of the heat rising but at a particular point the energy can't escape efficiently enough so it actually takes on the test the structure of you know hexagonal lattice so you get these little cells right and um if you've ever been to like the Pacific Northwest or other areas with like Basalt uh rock with these hexagonal columns you can actually see this exact thing embodied when lava cools and then cools and tries to cool release that energy in the same way and performs these giant hexagon hexagonal Rock structures um in any case my whole point of introducing that line of thought is that I think that there are with respect to you know homeodynamic systems uh this idea of closure is an interesting one because people are researchers such as the commodity or Kaufman as well with his work cycles and uh montavilla masu they have this idea of constraint closures which is why I went in this direction which is deeply tied into the question of in their idea of their conception uh they they use the concept of a of a constraint as sort of a symmetry across time um and so that symmetry across time uh think of something like an enzyme right the enzyme has its own structure that is maintained across time but in its interaction with some other process it sort of locally breaks that symmetry to produce a transformation and it sort of acts as a sort of you could look at it as like a like a functional or dynamical um like boundary that also changes what bounces off of it and then if you order these boundaries in a particular way you know and close that Loop which is where you get this closure which is in many ways the way that I I tend to use the word closure these days the reason I'm laying this groundwork is just because this idea of closure is very fundamental the kind of ways I've been thinking about this particular topic of emergency governance um if you know once that becomes closed uh then you can start talking about these patterns of Auto catalysis uh whether that's a self-reinforcing loop and what it actually self-reinforces uh and to the extent that we're talking about governance uh I think it's relevant because of the fact that at each one of those points right there are aspects there are side effects of that of that function that are the inputs of one of these other functions but then there are also side effects that can have uh other uh functions in the context in which disclosure might exist and I think when we're talking about governance and the the life cycle of governance we

might be talking about something like that initial closure coming into being functionally such that it's overall net byproducts might be adaptive but to the extent we become reflexive in understanding this is that conscious evolution is conscious management to the extent that we can identify the closure and start mapping it and actually and analyzing the points and the flows within our own minds as individuals we can start playing games with it to the extent that that closure and its side effects shift to a point where you know it actually shunts perhaps some of that energy that should be going to the Adaptive capacity of the overall closure into sort of sub games of people's individual political Dynamics personal lives wealth building Endeavors that are within their own frame of reference but not represented in that closure that they are now able to see and take advantage of right which you you might also introduce laterally and things with the cantion effect where it's like if you're close to the money spigot and you see the game that's being played you are more capable of taking literally that energy that's being released into the market and directing it into places that you know might be more to your own interests even if you are saying and claiming that they're in the general interest of that that broader Collective closure that we've created uh supposedly in the economic sense by the FED to to maintain homeostatic or homeodynamic stability at this this financial level so you know when you kind of bring in those those Concepts they're very resonant with the way I've been thinking about it but um I just wanted to sort of establish that context for Grammar with respect to closure because I think it's quite critical to the way I've also been thinking about these tendencies yeah that was great thank you with regard to that so the that last Point um when we're looking at the particular phenomenon or in human endeavor of governance we have a special case and I like I think this is very interesting to be able to focus so the general case we could talk about in the context of folks like uh tainter and uh [] and Hayek which is just things that happen in human-built kinds of organizational systems that have these various Dynamics built in governance is particular because it's of that sort however the whole point of governance it is in fact designed to be the thing that addresses the problematic you were just discussing right and so the whole point is actually the recursion or the microcosmic boundary where governance is is itself subject to its own uh and we'll call right now I guess I guess we'll use closure I'm not quite sure if I'm even using it right because that was a lot he was put in there but the uh sort of the failure conditions that we're talking about and of course you can then imagine that kind of an infinite regress of okay we'll do the government it's on governance and

then the governance on the governance on governance just is there uh reason to believe that there's actually a way to uh find a loopback whereby that that actually that whole meta system actually has something like a higher level of stability and and you and you raise them a lot of different points in the very beginning in terms of the three basic questions of what the word stability might even mean it might it might not mean that it lasts forever in fact rarely would it but it might mean that it fails really interestingly so that you know like like the biological organism I die I don't live forever but but my life might be part of a larger Arc that has a completely different continuity process and and that's actually the thing that we're thinking about the song I might die interestingly and so that's a very different solution to the basic problem right okay what else was in there there was something really really interesting where I kind of tried to focus okay so um I noticed also the the this uh the I'll call it geometry so almost like the the third mode so we discussed you discussed the um you know a characteristic and dissipative systems where relationships between two different structured possibility spaces um produces a new um architecture it actually produces order at a purely physical level and um and I would say this is something like there's there's principles of physics or principles of nature that are as fundamental as thermodynamics and because they're as fundamental they constrain the playing out of thermodynamics and it's important to recognize we actually have that that's in the toolkit we can use this and one of the things that I was thinking about is that that one of the things that oftentimes shows failure conditions in human Design Systems is vertical or horizontal asymmetry you know so for example if I have like one person endeavoring to govern but let's just use it actually in business that we actually the ordinary sort of business organization the notion of span of control in military and in and for all bureaucracies it's pretty well understood but if I have a manager or a business governance function um they generally can't actually govern more than a small number of other people right so if I have a single manager when governance functioning they're indifferent in governance say 500 000 people that asymmetry in terms of information like I think in terms of information Theory perception sense making uh and effective Choice making that can be actuated down will create a wide variety of gaps and niches for parasitism but maybe there's something to be found in the context of a proper geometry where the information asymmetry is inside of some sort of of conscious boundary like we came to think about in terms of what's actually possible in terms of the space of Consciousness how we can produce larger spaces of Consciousness and then how we can actually design information flow through within that space so that every [Music] every flow is actually held within within a governance envelope

that doesn't produce these niches for parasitism yeah yeah I mean 100 that's very resonant I think that in that context so so to pick apart this this is symmetry asymmetry question a bit more this relates to you know there's sort of a long line of uh or undercurrent of political science that's not that not as popular that sort of flows through you know like Machiavelli mosca Burnham you know that that vein of just pointing out the the side effects of this kind of asymmetry with respect to the idea of leadership or sort of the ratification of any system uh in so far as it must take place within a subset of the population it can't be like even the idea of democratic governance itself has to be acted out on the day-to-day by a fraction of the entire group uh on behalf of which is being played out in theory and so that that subset of people who are identified are then you know they become responsible for for enacting that game with a particular degree of integrity and I think that that concept of Integrity this is really this kind of relates to the maybe the quality of those governance envelopes you're talking about in the structure uh but also the scope and scale over which uh perhaps responsibility is one way of thinking about it but then also accountability because I think to some extent that accountability transparency game and to the extent more and more people are playing off out their lives uh while trusting you to do the right thing especially you know part of that's going to mean that more of their resources are falling into you with that trust which increases the incentives for you to potentially uh do something that they will not see you know while their eyes are directed elsewhere it's kind of like I like to use the game of you know if there's a the more money that's in the pot on the table right the more you have to be careful going to the bathroom during the game if everyone leaves the room and you just leave one guy in there uh you know you got to be a little bit careful because even if that's a very trustworthy person at that table you know it the larger that puck gets you know you start beginning to trust uh to test that that degree of integrity and that's kind of that that same asymmetry is like the better system of government the best systems of governments that create the more of the systems we create to generate value if they are actually successful it is exactly that value that's created by them that becomes the Temptation for defection and and and so this is very interesting with respect to this envelope question because it it transforms into this accountability and representation question I think uh and to the extent that we are forced to represent these systems or to have accountability with these systems I think we live in a very interesting time because we are beginning to be able to have uh you know representative granularity and transparency in ways that were never before possible even and then and using those to sort of act as proxies for maintaining integrity and binding constraints on people's behavior and it's an

interesting balance because we obviously you know if someone is a representative uh we want those people to enact the highest ideal in theory of their representative position uh that being said uh we also need to have that position be somewhat attractive and uh to the extent that anyone is is looking to act out that position if it is fully constrained and surveilled they might no longer be interested in taking on that position so you see you might have to you know one solution to that might be something along the lines of like well automate the whole thing but but humans don't like as embodied beings humans don't tend to respond well to the idea of an abstract entity that they don't really fully understand the internals and that kind of aren't actually hitting our mirror neurons in the right way either like we're not getting that reflective property of like yes I intuitively understand this thing I think we get also innately suspicious whenever the idea of some sort of abstract system uh you know acting as our representative or acting as our governance even if that were fully transparent and less likely to defect against us or provably would not defect against us like we might just rebel against it anyway so we there's sort of this possible Catch-22 there so so those are some I guess some thoughts there yeah that's interesting so oh that that one that last piece it's let me see how that's right the last piece which was called the AI governance piece it's kind of there's no kind of worthy it's tremendously worthy of a really serious Deep dive and maybe sooner rather than later right we're actually accelerating in the direction where that kind of Science Fiction narrative is um plausibly and let's not say like we don't know these things happen in Step functions they don't tend to go in a straight a curve uh but plausibly is on the Event Horizon so we may actually have to wrestle with that I would propose though I cannot prove it um with adequate rigor and so it would require a collaborative effort but I propose that is in fact that idea can't work and there's actually I believe an actual rigorous proof that is available that the notion of a fully automated government system is intrinsically unstable but that doesn't mean that we won't try to do it and trap ourselves thereby um so let me let me back up a little bit before we get there because we can talk about the stuff that's sort of in between and uh this is very much the tainter the tanter story more or less right so we get we get the idea of um so the first movement is is the relationship between contextualization and decontextualization um which may be the thing that humans specifically most fundamentally do is the ability to decontextualize we can pull the Salient elements of some particular landscape out as we see fit right not just um by virtue of evolutionary Fitness we can actually fluidly identify characteristics that we'd like to pop into the foreground and sort of reify down into principles and then most specifically Endeavor to govern I.E to optimize for or against or modulate

according to our own preferences um and of course one of the things that we can do with that power of decontextualizations we can then design architectures uh on the basis of that decontextualization which is the the term technology that's what that means right technology is a the ability to decontact use decontextualization to uh design in nature uh phenomena that are specifically optimizing for particular principles um that we've been able to pull out of of what is otherwise a larger a larger million um and I would propose that this is at the basis of both the strength and the weakness of the the design like the epoch of design and the limitations of decontextualization are very deeply woven into the fragility or the problematic of say tantric style curves the high all the things that we're talking about so uh you know of course in the process of decontextualization on the one hand we're able to identify What's called the most the most relevant or the most Salient characteristics of some system and then we can begin to control for them okay I want I want more calories per unit yield of wheat um but of course the the inverse is also True by virtue of decontextualization we move away from the ability to have high quality perception of all of the other things that are going on in that embedded system so we lose track of them and by virtue of losing track of them to the degree to which they matter and of course now we're getting into kind of like uh something around Heisenberg and deep chaos theory meaning tiny tiny differences microcosmic differences possibly at the plank limit tend to build up into large consequences so that as we decontextualize we render into the invisible we render out of the realm of design out of the realm of the conscious a whole bunch of things that were in fact by definition calculated via relevant relevance realization at the at the unconscious evolutionary level just by you know sheer actual having causal effect on the system and then of course we're setting ourselves up we're setting ourselves up to have unintended consequences Cascade out into our larger environment and then what we tend to do then this is the the tantric move the thing that I want to kind of bring into the into the foreground is that we tend to double down so what we do is we will briccollage or we will cluge new technologies at the edge oh okay this particular problem is showing up great I'm going to do the same move I'm going to look at that problem foreground or Decon texture realize the elements that I can identify in the problem put together a new technology connected to the previous technology we kind of get one of these um remember building mobiles when you're a kid you can kind of add something at the balance point and it can kind of layers layers down um and of course the the point that somebody like tainted very specifically makes and I think [] **also maybe not quite as clearly makes is that we then get lost in our own creation right the the machine that we have built is has so much**

complicatedness to it it has so much um so many different things going on that we're no longer actually able to do Master of our own machine and this moves us back into an unconscious design phase right so now we're effectively maintaining something that we don't understand in many cases Our Generations past its origin stories we've actually you know we have a tiny window of awareness in a much larger field that larger field doesn't have the advantage of a tightly coupled deeply nuanced complex fully contextualized characteristic instead it actually has lots of these little decontextualization gaps foregrounds and backgrounds but we're super not aware of it if you've lost track of it a lot and now this is part of the niche environment every one of those gaps becomes a niche for various forms of parasitism or become places where local entropy is collected and can't be addressed because they're effectively in various kinds of blind spots or in what what the Santa Fe people would call Frozen accidents of our own complicated Machinery I.E it's just a gap that that arose because the way that we built the control system and for us to actually just address the Gap we have to deconstruct the control system but the game theory doesn't work the cost of deconstructing the control system is something that we're not willing to do to so the local entropy just gathers and gathers and gathers until it festers and it eventually creates a breakage that we sort of unconsciously just accepted as what's going to happen okay I think that that sort of played out what I had there I can hear you by the way I'm just gonna get some more water okay yeah no problem no problem um yeah and and so I mean that that obviously ties into the dynamic of you know sort of at the more colloquial level the fact that uh that humans don't act in general creatures biological organisms but especially humans that are in theory through one lens known to be the kind of creature that attempts to look into the future The Horizon of possibility and and attempt to act in line with uh a manner in which that that feature uh remains habitable let's say uh but then there's always this long-term short-term tension with respect to you know at what point does the you know do the do the possible thresh do the potential threats start to uh realize themselves or actualize themselves at levels that generate sufficient uh sufficient you get to call entropy or suffering or you know something that sort of shakes you strongly enough to to actually or deeply enough to change Behavior or your frame of reference at the appropriate uh scale to produce transformation that uh can respond to whatever that blind spot that you're mentioning by the process of decontextualization uh whatever that whatever has been manifesting in that blind spot that is then threatening the Integrity of

the process or the the capacity of that process to to maintain continuity across time one thing that really struck me that's interesting as well and I think yeah I guess I've been playing around with a lot of these types of ideas lately in terms of uh with respect to words and the the sort of inversion capacities or cyclic capacities of taking a concept and and flipping it and seeing what happens this idea of decontextualization is fascinating to me because you know when you're discussing it uh in my mind something that's popping up is is another like way of looking at this very phenomenologically is that if you if we look at ourselves as categorizing beings attempting to attempting to and to me what categorizing beings tends to mean is is having a functional attractor of some sort that we're binding representations to with this sort of agglomerative tendency of more and more things that have that likeness or sameness or are associated with that network uh building over time um one of the the essential characteristics is that we can't focus on everything at a given point in time and therefore we have to live inside a given framework that mediates facilitates and and hones our attentional capacity to actually navigate that process of of using this category structure to relate to actualities and possibilities continuously across time and so interestingly enough the context of that ambiance and one way of looking at decontextualization might also be a process of hyper contextualization in the sense where we are taking a very small representative Subspace uh we're identifying a particular set of Concepts like you know I think abstraction is a way of looking at this right like you're saying there's a set of similarities here and they're graspable enough within my mind at the moment so I'm going to take those and then begin to act as if I'm living inside those as the full context right and so like through that zooming in to the abstraction or to that to that subset of the possible pattern space I begin to act as if that subset is the entire Ambiance and therefore become blind to everything that is not inside that Subspace and so you're sort of living inside the Micro Niche this micro gradient because you identified a useful gradient inside that abstraction for some uh some perhaps instrumental or or even you know even something that was truly convincing like um you know the idea of calculus is extremely compelling and we got a great deal of adaptive utility out of that concept but does it hit a limit at some level in terms of like the metaphors and the way of seeing the world through the lens of calculus does it start to break down I mean I would argue yes I think there there are boundaries when you start to see the world as a set of instance when with respect to my studies in Consciousness and and like the emergence of Consciousness and topology of

Consciousness I think one of the things that actually has done you know inhibits us quite a bit is trying to think of it as an instantaneous property of a system something that could in theory be pointed out as a as a a point on a tangent curve somewhere right like um so in any case like this recontextual decontextualization Frame I think is a very deep frame and and part of this phenomenological tendency we have to see patterns and then start living with inside those patterns as an Adaptive framework and then just to close this off I guess with respect to the coonian take on that as well you know in his he's well known for structural scientific revolutions but my favorite essay by [] is uh an essay called the essential tension which is like the essay in which she kind of responds to the public response to his structure of Scientific Revolution speaking in front of I think it was the APA or something like that at the time and you know saying I essentially have been largely misinterpreted by people who want to use that work as a justification for throwing off all constraints but in reality what he was saying this essential detention was like uh there's a frame that's established a very formal rigid frame and the point of that formal rigid frame and discipline is to move to a point of convergence not because of the convergence point is the transcendental ideal that we absolutely must get to but because by acting as if that convergence process is worth um worth moving toward as we converge we will realize what is outside of that cone of convergence so to speak it'll start showing up as aberrations and then by collecting those aberrant pieces of information we can begin to realize oh okay this is how we can print in a principled and structured and continuous manner uh once again open that frame back up recontextualize the decontextualization or or like Zoom back out of that hyper Focus contextualization um and and synthesize a new frame that doesn't just jettison the value of the previous frame uh arbitrarily but that also isn't myopically focused Beyond a point of critical uh uh uh critical collapse let's say um on pursuing that singular interpretation of the the old frame and so I think to some extent like this is the set of patterns that were that we're trying to work with when we talk about and we're we're trying to one I think we're we're instead of the structure scientific revolutions were like beginning to see you know is it we're beginning to see the traces of like the necessity of that analogous process occurring at the structural level of governance and and this this is sort of the Zeitgeist of our ERA with many people attempting to to place ideas Place new frame possibilities and which how should we direct our attention uh into the the space of public discussion and uh hopefully experimentation and try to figure out um can we get some positive signals so that as this prior frame begins to disintegrate um are there are there lifeboats that are actually capable of

carrying more than one to five people at a time yeah so um it's notable that that uh you know
tanner referred to it as the collapse of complex societies and um I think the proper way of
describing what [] was describing was not so much of paradigm shift as a paradigm
catastrophe and um I think there's a very specific point there we're actually looking for
Paradigm Shift you know it's like hey I got a paradigm okay now it's time to shift in that
movement and I I would like to very much evoke here the uh uh cognitive science
Notions of opponent processing and how the brain very specifically in you know all the
way down to the most basic animals uses modal distinction specifically what I was
thinking about is very much the dynamic we're okay where it's time to focus on a very
localized task but there's something happening at a meta level that allows and requires in
fact is Shifting back to diffuse and then moving back and forth right and that that notion
of modal distinction right brain left brain distinction are qualitatively different in how
they go about uh orienting towards where to allocate attention and even how to do um
sense making in the relationship to perception and then there's a meta Dynamic that
moves back and forth between them right so that kind of a tricameral structure um at a
very very deep modal distinction or qualitatively distinct level seems like it's uh the
proper design for any kind of new kind of collective intelligence and uh I was brought to
mind of the of the distinction between thinking and simulated thinking that I was trying
to make gosh about five years ago um and I think that was that was oftentimes also not
fully grasped so if you wouldn't mind I'll just recapitulate because the point is very
specifically the point that we have we have very we have two modes at least but definitely
two modes of thinking right one mode that I called habit mode one mode that I called
explore mode and I was I'm not necessarily trying to say that I invented any of this and
there are many people I'm sure have done a much better job I was trying to point out the
thing I'm trying to point out is not that um but the idea that habit mode like Paradigm is
able to take advantage of a variety of different Frameworks that are relatively well
designed for a particular domain and therefore quickly and accurately and predictably
be able to make sense and select appropriate choices in the context of a certain domain
right that's a very functional thing to have explore mode is a mode that is modally
distinct has a more diffuse awareness much more contextually aware is oriented towards
things like Insight I.E breaking existing assumptions and very very oriented towards the
subtle things that may not ordinarily be noticed in habit mode but may actually be
relevant and that's why we have to have this different Explorer right so the The Meta

process is okay which mode should I be in right now am I am in a context where Explorer mode is the most adaptive mode or am I a context for habit mode is the most adaptive mode and can I fluidly move back and forth right so that's that that third right so I've got one two three and the premise of this notion of simulated thinking is that there's something about habit mode where we can get stuck in it so it somehow habit mode breaks this third governance layer that allows us to fluidly move back and forth and then gets us more and more stuck in habit mode in particular where we begin to build a habit that simulates governance right and then and then we're in trouble because at that point we can't actually make proper use of explore mode because we actually haven't got the capacity all we can do is when the governance system is saying hey it's time to move to explore mode habit mode says oh I've got it and then produces something vaguely like explore mode but still within the fundamentals of habitat and you can you can sort of extend that to all the different examples we've used right this operates in every Dynamics you know so in Coon's model it would be anomalies are being thrown and the larger system is saying hey there's something wrong with our Paradigm but but the the paradigmatic mind says well I'll just produce a more complex version of my existing Paradigm to absorb those anomalies into it or I'll use the function of ignoring in other words I'll put them into a an area of local entropy and just kind of let them sit for however long maybe forever ideally right and the you know the meta premise is that's the problem right that is the problem of conscious design is that getting stuck and not actually having the opponent processing built robustly enough that we can move from habit into Explorer without having to go through a crisis right the problem is is not that we sometimes our habits aren't functional we have to move to a different mode the problem is sometimes we get addicted to our habits and the only we can get out of them is actually to go through hitting rock bottom and that's a very very un adaptive way of solving that kind of problem so yeah yeah and um and just to sort of like to test the language a bit and make sure that I'm I'm with you here and uh also use it potentially to synthesize some of the the topics that we've been talking about it's fascinating to consider this in light of of this question so in terms of parasitism that we've been talking about and also the question of uh the benefits of parasitism possibly right because you're talking about this this catastrophic or this catastrophe which is in you know dynamical systems in many ways analogous to to to falling off a cliff um like in the map that is actually exactly like that anyways um and so uh it could be it could be the

case right that one way of looking at parasitism if you can if you can endogenize it or if you can actually internalize some amount of it or look at it as a property of systems is potentially as a smoothing function for that Cliff insofar as you know this gradient that we were talking about before um in light of the language you're talking about in terms of uh this habit mode right and exploration mode uh with respect to Collective governance or Collective uh systems used for simulating this adaptive process as a collective um that Collective process of governance can at times be in that exploration mode potentially discovering some new form so if you you look at like the Inception of our you know the nation of the United States and there's a lot of synthesis happening a lot of different ideas you know from a you know lineage you know extending thousands of years into history uh drawing from all of that to try to synthesize a particular novel framework um it was very conscious hyper-conscious hypersynthetic very exploratory um but it created something that in its stability and efficacy became so functionally stable that more and more people were able to move into habit mode with respect to that as a foundation and as that happens uh interestingly enough it opens up uh little exploratory niches for actors to go in and say are there micro gradients in that system because of the fact that all these people are on habit mode no longer their attention is focused outwards it's no longer you know a 10 ending inwards and so they actually go in and and start siphoning energy perhaps uh in their own Explorer modes as perhaps parasitic Behavior Uh from this governance Loop but interestingly enough that can focus uh that can function as a way of um testing the stability of that Loop to some extent like they are uh forcing that fundamental functional closure of governance that people have stopped attending to on Moss generally uh they're forcing it to operate under additional constraints under additional load even though that load like that's an internal load that it might not have necessarily been formally designed to Bear but that parasitism you know like a parasite load on any animal is forcing that animal to operate um over and above what they would have to do otherwise without that parasitic load and so it's actually building some slack into the system perhaps such that when people when people when we start when things start getting shaky and people turn their attention back into that system see the parasites and potentially get rid of them there's still enough slack left in that mechanism hopefully to navigate uh the the cliff or you know to smooth the cliff so that's like a possibility I'm not really I'm not sure how coherent that was if that made sense I mean let me know but it maps to I believe pretty almost exactly to Brett Weinstein's um story about uh peacock

Tales of pyramids so his argument this is effectively peacock tails and pyramids are exactly what you're talking about the defective of the peacock Tails not just a pure sort of sexual display but it's actually effectively a parasitic Reservoir that provides a Fail-Safe that to the degree to which the context no longer supports it you can lose a lot of [] tail get back a huge amount of adaptive capacity without losing any real holistic um total uh Fitness and respond to what's happening same thing with pyramids like if you put a lot of social uh excess would that be called Surplus into pyramids as opposed to say for example more people your carrying capacity is actually buffered so that if there's actually a significant uh drought where food production is not as high as it was you just stop building as many pyramids and you don't actually have a cash problem collapse of the underlying deep structure was that in his thesis or I haven't heard him talk about that uh surprisingly I've heard him before so I may have just added them I apologize for some reason I thought it was like I was like yeah because um in the hunter-gatherers guide that one or yeah and uh maybe I I've haven't watched I don't know I need to reflect back on it I don't it didn't yeah it might be there but um yeah that's it precisely yeah I mean it's certainly not surprising to be like Brett and the mind that Brett has in the way that he thinks uh I think you and I tend to dovetail and converge to a lot of similar frames of reference so that's yeah but I mean I think precisely that these abilities to um maintain higher carrying capacities within any given system fundamentally provide slacking in that system and you know we look at it if we look at it just at the object level and and you know through almost entirely moral framework uh then uh it can it can it can lead us to believe that there's no benefit here and to perhaps be a little bit overly acerbic in our language toward what's going on interestingly enough it might have been selected for we might be selected precisely uh you know in the same way that it's we have some interesting side effects of psychopathy that might have uh overall adaptive side effects that small levels in society uh some element of sort of corruption and parasitism uh at a base level within even healthy systems might be optimal um to the extent we can talk about optimality in such a system and so um and that might and then knowing that might uh my bravest against becoming overly cynical right um because I think that at some levels the at some level the the greatest threat to our current capacity for adaptation uh is a abundance of cynicism uh and an abundance uh and like innervating quality of that cynicism with respect to to sapping us of the the necessary will energy and belief that that we can actually do anything that is uh that is worth doing so nice Daniel how much time do we have left 25 minutes but I'm happy to stay on anytime you want to feel free to continue this section and then we can go into q a I have

another piece that I'd love to drop in right here um so let's see yeah the psychological consequences of being stuck almost like the psychosocial consequences like the feedback loop so on the one hand you have the interior psychology of being in something like simulated thinking um where you can't get out of your own habits of thought and you can't escape into a higher level of insight you can't escape back into explore mode but also the feedback loops are being stuck in a collapse and complex society where uh explore mode is in fact inhibited because it doesn't have any place to get energy or resource for example or positive feedback the energy resource and all the different human needs modalities and so what happens when you're in there and what happens there is you start getting things like you get depressed um you get cynical you get nihilistic you get anemia right okay so let me think okay I'll do it three things first that notion of parasitism as a as a reservoir of potential is very interesting and I can think about that immediately in two ways two examples that pop up uh one is that the notion of like eighty percent 85 of people are in so-called [] **jobs or at least self-reported um which I think is very very reasonable particularly if you sort of think about the Cascade effects um meaning if you were to remove a whole category of [] jobs you would realize there's a whole other set of jobs that are actually [] jobs produced entirely by the first category so if you start unwinding them well on the one hand that's terrible right I mean people are wasting their lives and their energy and just spinning on the other hand oh that's really good because that actually means that I've got sort of 80 of the human population is sitting on the sideline waiting to be deployed generatively as soon as we can get our head out of our ass right that's nice to catch both ways um I also recall during the the 2008-2010 financial crisis I'm saying okay well how far down does this system actually call it need to fall for truly robust reset I.E if the too big to fail systems have been allowed to fail properly which is what should have happened what would that have looked like it's like well we're probably not going to drop below like a 1972 level of standard of living like that's a pretty big fall from where we are right now and I recall the 1972 was actually pretty nice and it wasn't a terrible style of life and the point being that there's almost something like the uh what are they called the like the F-350 hyper truck well it turns out that if if we have to actually give up our F350 and just scale back down to an F-150 oh that's actually not that bad we'll survive that's not that's not a catastrophe right that's actually a pretty decent amount of fat built into the system that we can allow to cut back and not lose our [] now the challenge of course is can we do that elegantly so then I'll get to the third piece um everything that I'm about to say is from my**

opponent is entirely hypothetical I am making up this story on the basis of some some heuristics but the the notion is something like this kind of dynamic of um modal uh uh opponent processing and some kind of system that that catalyzes the movement back and forth between them is is very very old like deeply fundamental it's too important not to be woven in like the obviously the sympathetic sympathetic nervous system um go way back all right so that basic structure is super fundamental and shows up all over the place all right second humans at the at the kind of the first apoc indigenous humans almost certainly therefore would have used that at the social level right we would not have been able to achieve something like a social cultural style of collective intelligence that didn't also have this kind of structure woven and deeply but it would have been woven in deeply using embodied cognition obviously not design cognition because we weren't at the design recognition apoc so what does that mean what does it look like the proposition is something like when the external context when the conditions that we're in are signaling a need to move to a group explore mode exiting paradigmatic obligate paradigmatic mind or stuckness exiting habit mode into a more liminal form of collective intelligence there's a every distinct human at an embodied level will sense that differentially and they will subtly begin to signal that differentially back into the collective environment I'll give that concrete and so it's been raining less and less we don't have concepts of of rainfall we don't have meteorology like there's a long [] **time ago but we notice in a very fundamental sense our bodies are a little bit more dehydrated or the energy necessary to get water a little bit higher we're noticing the prey animals are reduct are not as common we're noticing that uh yield of food is decreasing noticing at the physical level deep deep stuff like biological and then there's different types of humans right we have a different typology we have old women we have young women we have young men we have children we have older men right there's all and probably I'm guessing something on the order of like 7 to 40 cognitive typologies that cover the different um kind of modes of reality that it's useful to have strengths and weaknesses in it right so this is a there's a collective intelligence that the indigenous level is constructive typological differences each of those typological differences will have differentials and sensitivity to the subtle signals of is it time for us to actually move to a new Niche do we need to leave the valley and go into the mountains because the rainfall is telling us that flood time is about to come that kind of thing right and it will show up in Behavior that's the key thing like actual physical behavior um physical Behavior will produce a second order feedback loop right as I notice the older women behaving in a different way it**

signals me in an embodied level to begin and the ACT adds additional differential signal to my own perception and then we'll change my behavior and so when I'm noticing for example that the young women are giving attention and perhaps sexual access to a different kind of behavior in men very clearly this will change the behavior in young men but that also will cause to tend to Cascade throughout the larger environment so the idea is there's something like a a very very complex set of multi-typological feedback loops pulling in subtle information from the context that will eventually catalyze a pivot from tribal paradigmatic locked in very functional and effective adaptive habit mode into okay let's all sit around the fire hand off the comp from the chief to the shaman let's talk about the oldest deepest mythological stories hey Elders are you aware of anything like a very different mode right um the premise I'm making is that that's actually the case I'm going to argue that we have that that a large part of what's actually going on like characteristics like depression and cynicism are feel like frustration you know how frustration works right frustration is when there's a deeply felt sense that there needs to be some change in the world that is building up energy to make the change but the change can't the path to the change can't actually be produced and the energy effectively feeds back on itself right that's the energy of frustration which eventually will either build to a release this way or it will uh come to a feed it'll cut itself up to the source and then it will lead to more like depression right so that's a proposed like to put out there that we're in a moment a big part of what's going on is that we're all receiving the signal that we need to shift into this um Collective Consciousness mode this liminality mode this this group explore mode we don't know how and we're largely stuck and so we're trying to figure out how to make that move at very micro level like there's lots and lots of adaptive stuff going on not the least of which is the Brokenness of our body mind our body is in our mind in relationship to just reality ideology overriding our signals like in body cognition is way out of whack um and a big big chunk maybe even like 90 of the chunk of the problem they're actually dealing with is the Journey of Simply restoring ourselves to that basic capacity once we get back to that basic capacity unlocking the 80 of people who are sitting in [] jobs back into a coherent collective intelligence that's operating in a group explore mode has enough capacity to actually navigate us to the niche we want to get to so that's a I've taken the problem of governance and say okay that's actually we can take a series of steps and think about how do we maybe find a way to work our our path how do we find the path to get us back to the location where we regenerate our ambient

indigenous capacity to operate in this mode from which we should unlock a vastly larger capacity to think about these bigger problems right and the end of point yeah I mean there's there's a lot there I think I you know I could go I could go on a I I'm quite the response I don't I think in terms of time though I I think Daniel you did want to bring in some live chat questions as well um or in terms of where do we want to go here because I could I could respond to that there's a lot of really interesting points there uh or we could weave questions into that uh that context that frame um where do you want to go Matthew with a direct response then um we'll have a rapid question and answer that'll take us just around the time that we agreed upon and feel free to depart as well and then at the very end I'll just make some Reflections and connect it to active inference so first Matthew then question and then feel free to stay or leave and they'll be a little active reflection cool yeah and in light of that I was also really wanting to raise some of the uh the active inference uh semantics and context in response to in response to uh Jordan's points um one interesting frame that he'll it's interesting to observe that it hasn't saturated the culture perhaps just given the mimetics of when and where is introduced but um sort of like EO Wilson and uh Edward Lumsden introduced something called dual inheritance Theory uh 20 to 30 years ago I'm not sure if you you encountered that they had this book called uh genes mind and culture and um there's a lot of residents with respect to the mechanisms you know they did a lot of work with respect to the manner in which the kind of uh diversity of the behavioral sensitivities um psychological psychomomatic and like cultural level sensitivities and behavioral patterns uh have these kind of emergent feedback loops that uh that essentially end up creating a emergent form of parametrization at the cultural level and the cultural Dynamics you know these changes you're talking about at deeper levels of reality as they bubble up through and restructure the cultural manifestations and the sensitivities uh and behavioral Tendencies of this diversity of psychotypes I guess you you were saying how that actually as a as a mechanism feeds back into uh and back down through all those other different layers of adaptive uh mediation selection transformation um and sort of like this this emergent Cascade and then this essentially reflects the parameters parameterization which many people have struggled with in the fat in the past is under the offices of the argument around downward causation that's a whole slightly different conversation but I do think active inference is very interesting from the perspective at least from my perspective some of the work I've been doing recently showing that in many ways uh active inference is in it offers a one one function in the category the necessary category of transformational mappings between perception and action that do

exist at this deeply fundamental level uh in terms of the the sort of establishment of any causal closures either you know you know of any non-trivially complex uh process any any cycle that's capable of observation and action based on perception right in relation to perception um and interestingly I think that that does biasure nature in the way that it connects to itself give rise to this bimodal topology like through a particular kind of non-orientability and so the reflectivity when something comes back to you has a different orientation so you actually have an understanding of um you know the relationship between past and present via the lens of duration because of the need to perceive and then impose some transformation on the environment through that I think we're just beginning to develop the the capacity to work with those tools in a way that doesn't uh impose that that's more like surfing these adaptive waves that allows us to observe the way in which these Transformations and perceptions are mapping onto actions and using that as a tool and and a potential in a toolkit of like explicitly and consciously designing networked connections between these many different perceivers and actors and their perceptual and action capacities in a way that can um I don't want to say provably but but but prince in a principled manner give rise to emergent systems that do have these properties built in endogenously and perhaps these signaling mechanisms built in endogenously in ways that can resonate with what you're talking about which is the embodied sense that we have intuitively or unconsciously that arises but we don't necessarily have the language or the framework to explicitly talk about it so sort of that that that emergent welling up of emotion or frustration or depression or enemy as you said gets shunted into Language and Cultural symbols and representations that don't fully close back on the on the sort of actuation capacities that we need to actuate or that we need to upregulate in this particular moment and I think active inference and and other uh Frameworks like it precisely because they take this perspective epistemic perspective that says actually you don't get to directly observe the world all you can do and the best you can do is take this humble process of of trying the best you can to connect signals of perception to signals of action in relationship to preferences at a given time and the context in which they exist and and build from there uh ground up I think it's it's a leading candidate in the the modes of phenomenologically Shifting our perspective through um you know through new ways of operating and seeing the world through new paradigms through uh new toolkits so you know that's why I think I've been you know really interested in the active inference space and see it as as something that's quite relevant to this entire conversation and in particular to to your sort of final remarks there all right excellent awesome thanks both for the dialogue this is really great so uh I will try to

bring in many of the questions that were asked in the very active and energetic live chat will get pretty quick thoughts that will conclude the primary section of the stream and then if you want to stay or leave but I'll just add a few more points because indeed the connection with active inference is very Salient okay so into the questions we go first from Charles Bloss hello what are the primary patterns including values of wise collaboration one was the one that you just noticed which is moving at the pace of wise collaboration meaning in this case I noticed that's a very complex question I'll have to take a lot of time to think about it yeah questions raised yeah we raise the questions here and explore it in the coming time but feel free to give a short response while you can for sure for sure I mean it's just interesting I think there's a continent there because one of the things that was coming to mind as I was taking the time to try to reflect on that was was patience hey especially in light of all of the tendencies that shift us away from the capacity to take in sufficient information relation uh uh to cycle on that in in a uh to to so there's a tendency to always want to especially under pressure under stress to always want to take perception and immediately map to action right like for every cycle of perception some sort of symmetric action because it feels very it feels uh like you're at least doing something feels better than doing nothing but uh there might be some minimal necessary amount of reflection you might need to cycle a number of times uh at a number of scales with a diversity of opinions before the capacity is built up in that Collective um intelligence or Collective coherent structure before any action that it takes externally can do anything other than make things worse and I think that that's a massive problem that we're confronting right now because most of our tools are designed to map perception to some sort of extrinsic action almost instantaneously and we haven't created the infrastructure that allows us to more deeply uh allow for complex emergent perception and mediated mediated in the context of action at the sort of duration and depth required to meet the moment great so then a follow-up comment and then a question so Charles followed speaking of time flagging an inquiry in regard to timekeeping and calendars and their specific numerics effects on Humanity's psyche and cultures related to this patience and Kronos and Kairos um next question from Nathan Curtis at what point in time from instantiation does the established government lose its ability to be self-referential and able to look in on itself to identify flaws I've had it I've had a general thought about this question um what I would say of course is that the answer is is it depends right so the particulars matter um and I think it's going to be something like three primary characteristics one isn't going to actually have to do with the let's call the shape or the topology of the underlying architecture so an architecture like say

liberalism has a decent number of uh designed fail-saves very specifically conscious consciously aware of this problem and so it has a it's more like a dissipative structure it has a higher capacity to actually live longer given a particular gradient we're using the earliest language we were using the second it's funny you can actually you can basically identify three very straightforward principles and one is the the amount of of energy or the amount of of information or or potential that was enclosed in the initial uh design the second is the the intelligence or Elegance of design in terms of its actual uh ability to produce structure that most effectively navigates that that dissipation into entropy the third and this is I think a piece that we're running into a lot is something like a characteristic of bandwidths distinct from time meaning the proper measure is not chronological time Chronos but it's something like relational interactions or Loops of perception to agency something like that and so if I think Jose is a kind of a clock that is the right clock awesome yeah I mean that that Maps directly onto you know kind of what I was mentioning with respect to the active inference idea of duration as time right in the sense that that is precisely that cyclic relation in terms of perceiving the consequences of action as perception and then you know internally inside of whatever mechanism is perceiving and acting at whatever scale we're talking about um you know that transformation is sort of this this this era of time that we perceive as as the linear structure that linear mapping but the real time in a lot of ways the systemic time is the the path links through this causal structure that flows um between the action of Any Given entity and uh the the capacity to perceive feedback uh in terms of what those actions caused and how that allows you know in light of the contextualized relations that give meaning to the action in light of the values that are ostensibly pursued and so in getting back to the question in terms of when is this system capable of or when does it become no longer capable of of that kind of self-reflection well I would say to the extent that that cyclic time that capacity that closure for being able to understand the causal effects of one's own actions in a contextualized meaning grounded frame to the extent that that is intercepted or broken right or to the extent that it becomes no longer concerned with its own or solipsistic is another way of putting it so those would be conditions right it's like to the extent that it actually uh becomes so trapped in its own existing frame or its own assumptions of causality um so this would be you know in inactive inference you know you have something along the lines of um you have something along the lines of like an over um like the Precision of its its capacity to represent the world becomes far too precise with respect to the the the the relationship between a very narrow subset of modalities and factors in the world percepts and actions in the world which is exactly

what we were talking about earlier when we were talking about in Jordan's language the decontextualization or like hyper contextualization um and then you know that prevents you from being able to identify um a very large percentage of the causal structure that might be relevant to whatever is threatening the coherence of your collective intelligence or your actual physically embodied metabolic systems or or whatever that may be so yeah awesome Nathan bro if we want those representatives to embody a certain aspect of being in that position how do we best select those people what modalities can we use to select the best AI recommendation systems temperament and character inventory test I'm a person I'm fascinated by the possibilities of uh emergent identification um of of reputation like networks that take advantage of of the information that exists in local relationships uh we might notice that to the extent that we have elections at larger and larger scales we lose that right like in your local elections in a democratic election or Democratic electorate um you know if you're if you're voting for uh for your county like you know for for sort of like a county chair um or a uh Governor or whatnot you're much more likely to have relationships uh or at least some sort of socially contextualized understanding of of who that person is and what the words that are coming out of their mouth actually mean in relationship to your embodied life um but the further you get away from that the more you're dealing with abstractions and so the more energy has to be put into making those abstractions feel real to you and then you enter the world of you know television news and political uh news Cycles very myopically focused hyper precise a lot of energy and time put into those particular issues that get the most attention as opposed to those issues that tend to actually have a more embodied effect on on you know an individual so how do you actually take advantage or Surface that that emergent pattern of contextualized wisdom not just within Democratic elections but at at a much deeper level than we've ever considered before because I think One requirement of meeting the level of complexity of the problems before us is representing a much greater degree of information which without that information uh you know and that information being at many scales of our net emergent organism being the nation state let's say for now um without that information flow through the system overwhelming the system I think liquid democracy perspectives as a general category are quite interesting in terms of looking at them as as sort of like graph finding or pathfinding algorithms of expertise that are mediated by local expertise or local relationships local trust networks trick with those is is I think executing those in a way that is um that is capture resistant that doesn't necessarily make that doesn't turn the entire game of life into a game of of politics such that uh we are all campaigning all the time to be perceived

as the greatest expert in whatever domain without actually producing the actions that would naturally give rise to those perceptions it's a big problem it's like aping aping the taping the behaviors that would lead to someone perceiving you as successful it's like the Instagram model or Instagram personality problem right it's like if you rent the Jet Plane and you know put on the facade and you wear the skin suit and have the persona it's increasingly difficult for a large percentage of people to divorce that from the real signal itself so I think within liquid democracy uh patterns of sort of emergent stewardship or identification of people who might be able to solve problems with greater competence that emerges as a problem and then there's also the problem of response time so it's like to some extent uh the reason why we have Central executives are the reason why we have individual representatives and place Authority in the hands of single individuals is because there are particular kinds of events that require response that is at you know at the scale of smaller time Windows than is typically capable with um processes that include large numbers of people so that's another attention uh in terms of needing to solve that without uh without falling back into sort of the central executive idea but those are just sort of the cluster of thoughts that come into my mind when that question arises um what the first thing that come up for me is that I would I would profoundly caution that principle of asymmetry that I'm reasonably confident that the right way to do this is to almost perfectly invert our current way of doing things and our habits and intuitions meaning um a radical regeneration of actual real relationality and orientation of sovereignty I.E Choice making Authority and power to the lower level more intimate locations is profoundly the right way of doing things and so that most people for the most most of the time most of their choices are actually being governed by high context Dynamics um and then on top of that substrate you then can actually relatively easily build a construct that allows kind of a distributed Cincinnatus where when it is in fact very important to identify individuals of extremely unique capacity to deal with very specific problems that aren't available within even a vaguely large local environment you can press them into it but the point is that they're so deeply connected 80 of the time into a actual lived context with all the different um how does this like strong bond High feedback loop high quality subtlety that the capture into the Instagram model thing it doesn't really have it's obvious that that's a really bad idea right if you have people who are living almost entirely in Virtual lives where there's really nothing real in their lives that they don't actually have real friends they don't have real territory they don't have real relationships with even like their parents or their siblings then it's very difficult for them to notice that the Instagram influencer thing is terrible right because they don't really

have something other than that that is qualitatively distinct but if they're actually living a truly healthy life with all kinds of good positive things happening then you can pop into this other role notice that you really don't want to live there do your job and get the hell back into a nurturing environment um I should mentioned by the way it's not clear to me that we can actually get from here to there air outside of catastrophe right the navigation of that particular path um at the very least is a serious serious serious job clearly well beyond the capacities of anything that we have right now so it would be kind of a Tropic Cascade of a series of upgraded capacities that get us to the point where we have that capacity and you know that's uh yeah good luck that was also I think we've talked a little bit about this or exchange a little bit about this uh like on Twitter in terms of one of one of those constraints and when you're talking I just kind of wanted to get your your take on this one of those issues with respect to potentially walking that adaptive uh that adaptive Valley right or like trying to balance upon that like adaptive bridge line between our current local Maxima and perhaps some sort of you know foothold on uh on a on a larger Fitness landscape um or higher point on that Fitness landscape is this idea of national capacity in relation to the geopolitical structure of the rest of the world right because like we're operating not in isolation we're operating you know if we if we delegate uh if we delegate responsibility like this transformation that brings agency outside of that Central locus of coherence that is now this this Behemoth The Leviathan of the state with its um primary focus let's say or at least it's ostensible reason for being as the maintenance of the integrity and the boundaries of the entity called the United States or you know whatever the nation might be at that scale if that you know as soon as that sort of starts to falter it's also perceived by other agents on the landscape as opportunity um and then interestingly enough is it possible to maintain that transformative traversal in light of whatever pressures are added on top of our already great list of pressures uh by other geopolitical actors and I I suspect that that's at least one possible reason why you think that you know a transformation like this will have to be forced out of sort of a terrific Cascade collapse or you know cascading collapses um but yeah I'd be interested to hear your take on on that pyramids so by hypothesis um let's say that the American Empire is one giant peacock town and so we're going to lose it but the good news is we can't we can afford it in fact we better off without it um and so yes I think you're right the the and it's funny it's almost like a expanding to fill a niche right we found ourselves in this odd circumstance of being first a uh one of two superpowers and then being an actual Global hegemon for the very first time in human history and we expanded to fill that Niche meaning we we sort of depleted all of our our excess

capacity until we sort of uh I'd say unconsciously and arbitrarily sort of deployed the global hegemon game all the way out to the edge right they were sort of pushing at the absolute limits of our control capacity uh and by the way in the context of all these other things we've been talking about like local entropy and parasitism et cetera and all the niches would open up in that interior and I would simply propose that we're already past that point so uh one might argue that any kind of change of behavior it comes with cost of having to give up the benefit of being a global hedge fund and maybe you don't want to do that that's kind of the taintor uh pivot point if you're familiar with that point in the in the arc um how I think that dye has been cast that we're right now in the process of a forced uh Cascade down from that position into some new multipolarity and that can be and I suspect we've probably gotten the order of like three to five maybe seven more years for that to kind of land all the way down the land very clearly that that's where we are but I would say we're past the top of that point so it's a downhill let's roll from there and much like this notion of [] **jobs or the the demoralization cynicism of being stuck uh we might find ourselves demoralized by being unrooted from being Global hedge fund but we might also recognize that was actually never a thing that ever that we really really aspired to be and could have been or should have been more importantly could have been it was doomed in from the very beginning and that we're right now being forcibly liberated from a highly dysfunctional location which then if you sort of shrink the scope back uh down to a much smaller scope may actually produce a large enough um large x of of time and control capacity to afford US called The Generation a generation and a half to re-establish a more generative governance infrastructure so that's a an interesting possibility at least it's a nice story wow okay we're gonna live a path associated with that at least it's not that path something interesting okay to ask a few more questions just feel free to make sure I have to be very careful with timing because my wife's gonna be coming back with my four-year-old and I have a immediate responsibility so we can we can go but be mindful I mean literally hit the red button and be gone sounds good please depart um as family demands so just I'll read a few more questions and then again I'll just stay after to give a few Reflections so what is a good example of a major expanded decontextualization that we live in that exists in the culture now expanded I didn't understand that a major expanded decontextualization maybe a broad scale okay okay maybe that might be like a synthesis of the the two ways that we were talking about this and maybe in terms of expanded decontextualization and sense of the the hyper so the the tendency to see the world uh**

maladaptively um through a lens that has grown so ah okay so large that we can see nothing else perhaps the code policy yeah covet policy is a great example of exactly that right we we built a a medical and um Healthcare what is it called Public Health infrastructure pulling forward decontextualizing a very large number of aspects of of about embodied embedded humans and effectively with that digits that put a series of knobs in the cybernetic control structure that when a real problem came up that has extreme systemic complexity the way we ended up dealing with it is we started pulling these knobs as opposed to addressing the underlying systemic complexity we were like okay lockdown why lockdown well lockdown may be an optimal strategy if you think that you could shut down the spreading of a virus and are completely ignoring all the other systemic consequences of lockdown for example it was kind of like a you go to war with the Army you have philosophy except that the control board we had was extremely low resolution and even contraindicated by our own past advice in many ways but but it was at hand and so we pulled the levers um even though there were plenty of us who were saying you know perhaps treating this as a giant light switch is uh is not wise um but isn't always the best play yeah yeah and I would also just extend that point into the the fact that beginning to reflect on that as a society is also showing us that there are a number of other uh decontextualization or or sort of like frame lock issues that we have that are quite uh that have built quite up in a lot of inertia around them especially around the domain of what we call science or what is now parodied as the science or uh scientism which is like the slightly the more serious uh conversation that's been going on since the Advent of uh essentially early post-modern philosophy in the van sort of phenomenology like who Cyril and who Cyril's questioning of uh of of sort of the the paradigmatic uh nature of science as a way of seeing the world that can displace all of our ways of seeing the world if we're not careful and um the tendency of that to because it has provided such a again this is sort of this this question of um exploration finding uh capacity and then that capacity to becoming so powerful that it it puts people into habitual mode and that habitual mode becomes a fallback if other levels of social uh or psychological or even mythopoetic coherence break down so you know to the extent that people lose other frames for perceiving reality uh perhaps for for interpreting their phenomenological meaning which is you know what the mythical poetic frame has provided for so long uh to the extent that breaks down sometimes for good reason I mean again there's a lot of corruption in the church and in religion and there's a lot of I would

say decay in the sort of interpretive frames presented uh to um to upcoming Generations I mean I was one of those people raised Catholic um initially left the church became essentially anti-theist for quite some period of time before beginning to realize only through evolutionary education and looking at sort of the game theoretic lens in terms of deep history that you know there was something there there was a there there and that it wasn't necessarily something we could jettison but many people have I think many people were in that same position and being in that position have jettisoned in that frame falling back upon the purely uh scientific frame but not themselves doing science being associated with the process therefore falling into a dogma and being willing to trust the high priest therefore creating an incentive for high priests who are mouthing the words and playing the game of Politics as opposed to really committed to the continued process of discovery and transformation and representation that we know is science and uh you know getting ourselves into that whole situation and I think we're still a very live conversation we're still trying to figure this out and you know what is What is the relationship between science can you have a scientific exploration of mythopolitic concert trucks I think that's a really fascinating possible field that also active difference would have a lot to say in terms of the emergent structure of collective Behavior Uh that's sort of the same kind of questions like really digging into and beginning to examine the kind of questions you were talking about earlier with respect to what it means for different people to have different traits Tendencies of perception personality traits what that ontology looks like you're saying maybe 7 to 40 uh psychotypes or something like that what the Dynamics of that as an emergency system actually look like in terms of their feedback and parametrization on the underlying structure and evolution of our society and how mythogletic structures and scientific perspectives change those perceptions and interaction with one another in Dynamic feedback mechanisms with one another or dynamic feedback relations with one another I mean these are all questions for the next 50 years right I mean yeah very important ones I would say yeah so I I liked it for this we're listening I like the fact that you were actually in describing a decontextualized uh camera the structure but it's describing the structure you also were articulating where a lot of the failure conditions emerge and how we find ourselves um like this systemic shift like you don't notice that you're actually moving from science into politics or moving from science into theology into politics and because you don't even notice there's a thing that could happen it's not a risk that you're even contemplating in your behavior and so

you're finding yourself on that on that trajectory maybe the the mother of all like who actually met a point being where we are like nearing the end of something the the short answer is to the question is kind of everything so everywhere you look is a hypertrophied example of extended decontextualization and so the problem is actually being aware of what the opposite might even look like um but maybe the mother of all is education the education system I would say that the um the current education system decontextualizes humans it's a development developing humans so radically as to deprive them of probably something between 70 and 90 percent of Developmental needs which produces probably a net yield of probably a 99 reduction in total human capacity um and many of the things that that Matthew was just bringing up like the embodied facility not just not just the cognitive scientific awareness of the particular psychotypes and their relationships but actually building actual facility in navigating that properly how you individually navigate the world and how you can be in right relationship with distinct other kinds of humans in collaborative spaces like that's actually really hard and it's not the kind of thing that's easy to learn when you're 50 right it's the proper thing to be learning between the ages of like four and twelve and then deployed effectively in particular kinds of contexts between say 12 and 20 like so um sorry that's a I'm doing the same thing I'm pointing out the things we don't even notice that we have lost my virtue of becoming hyper hyper focused on a very particular set of optimization and also education as I think a very very big example of the of the basic problem yeah and I think that that form of Education to the extent that we can do that we can also begin to see you know in light of what you're saying with respect to uh it's almost everything at this point a key element of that almost everything is the economic domain in which this plays out uh you can consider that sort of like a uh hypertrophic exploitation of supernormal stimuli right like what is a supernormal stimuli like these stimuli that we have that are essentially we've discovered as part of this reflective process of Discovery in the domain of science we've discovered particular ways of tapping into evolved um evolve perception action Loops that are unbounded in their sensitivity right like there's no you know it's very hard to identify upper upper Bound for uh you know how the extent to which you can advertise self-undermining uh uh food products or you know ways of spending your time if in the moment those produce some sort of extremely positive or extremely uh uh some high salience positive affect uh uh experience like eating sugar or um you know getting that dopamine release from the anticipate anticipation of acknowledgment of

one's perspective online or you know these types of economic threads that we have created are specifically you know the profit maximization threat I think this is this is often put on the shoulders of the word capitalism uh as a pointer to some sort of hand wavy like there's something going on here and it's capitalism um but I think at a deeper level what we're pointing to here is the fact fact that because of this very long lineage of frames that we have chosen and and then in many ways like adapted to live within as a function of past adaptive challenges we now live inside this maximal representation of uh of those ideas in a way that is no longer sustainable so like profit maximization has a myopic Focus leading to identifying those narrow set of Dimensions that are most in line with that narrow metric over the sort of corporate function and then tapping in that that actually successfully tapping into aspects of all the aspects of human psychology and perception that are unbounded it was a search function and eventually it was going to come to rest on like probing The Human Experience in mind until it found those parts that literally would not uh you gotta go I gotta go guys sorry indeed Jordan great talk thank you in any case yeah that's another that's another um that economic threat is another thread to pull on but that was a great conversation in this uh in the active over time I just want to run through a few hits yeah that I uh wrote down from this great convo and just plant some seeds for how maybe we can continue 35.2 or in the future all right so in the very beginning there was a discussion of something like a meta Theory among the three of [] with respect to science heck with respect to business and tainted with respect to politics and we can even see that meta Theory as the prototypical perception cognition action or eco-evo Devo Loop so you spoke to the transformational mapping between perception and action which is basically what cognition is we just say it's everything between that map between perception and action and hence active inference as a prototypical framework for not solving but framing some of these contexts then yeah there's a structured way of mapping those first steps onto available actions and then and identifying I mean interestingly enough like one of the most essential aspects of the framework is that capacity to switch between the development or like reduction of variational free energy in the moment understanding like whether my representation of parsing reality in action what I'm doing is actually in line with uh the causal structure of my local environment versus this explore exploit future oriented frame of reference in light of um when is it time to actually uh search out some disruption or some entropy so that that entropy injection into my locally uh ossified frame can open me up again to uh navigating that adaptive landscape and introducing new

maps between perception and action that can facilitate my continued adaptation awesome so I wanted to actually go to the free energy with we had the earliest Epoch of human collaboration which Jordan characterized as sort of an unconscious and that is a lot like type 1 thinking in the type 1 type 2 system and it's a lot like variational free energy it's just this real-time unfolding ball goes downhill modality type 2 thinking second stage civilization as per Jordan and expected free energy is where we actually have the double-edged sword of the rollout and anticipation and we've seen simulations in octave where a tiny proportion of hypothesized negative outcomes even if they're totally avoidable Drive rumination and anxiety and high variance estimates and then the question is is there a third Epoch is there a third mode that uses attention to modulate the balance or the Reliance between the variational and the expected free energy and to what extent is that attentional modulation implicit tacit on board not automatable and to what extent could that kind of wisdom be reflected in our externalized niche and that connects to the question of Automation and transparency around our information architecture Communications and our infrastructure I I thought of first a sort of infrastructural incompleteness or instability theorem like even if you get all of layer one automated then of course Layer Two has not been automated so there's always the hyper prior and um also that Min to opponent processing thinking with tensegrity architecture compression and tension could we have systems where instead of just trying to oh we want this object hanging here we'll just put a pillar underneath it and then we'll put a pill underneath that like are we going to compress all the way down or could we have a harmonic mode that involved compression and tension and that's how it stayed flying then just yeah last oh go ahead and then I have two last points I I'd go either way but yeah I mean there's there's a lot of resonance here with some of the things I'm working on literally right now in terms of uh attempting to uh attempting to work with active inference as a like I was saying earlier as a uh that cognitive mapping like that that mapping you can look at it also as a category theoretic mapping um within a category and a set of morphisms right between this sort of uh space of perception and the space of action and you can then also in that sense integrate that into a larger hypothesis around the topology of emergence and then develop that as a toolkit which is interesting because inactive inference through the lens of active inference in theory everything can be active inference and then it leads to this sort of recursive hierarchical structure but following thinkers such as like Robert Rosen in terms of his and like uh vashevsky and relational biology and bringing in theories of a temporal causal structure that basically can contextualize the temporal unfolding of entailments you can actually use those understandings

to you to to take active inference as a as a tool and weave it into larger Fabrics like you're talking about and and those fabric connections they are related to active inference but they're not necessarily active inference itself although you could then take a meta step outside of that and say we can apply the principles to the entire system but but there's this role of understanding novel closures of observing of reflexivity and that reflexivity I think is a really like the person I describe is the approach I'm taking to introduce a a different kind of reflexivity and a kind that I believe is consonant with multi-scale arbitrarily complex emergence into the to the world of active inference but you know experimenting with how to stitch active inference together into itself I think is a very uh you know the entire field is new and exploratory but even amongst that it's a is that the sort of periphery in the edge of exploration indeed and just these last two points uh Brie collage was mentioned and active inference helps us with the composability legibility and simulation emulation analytics on different systems physical cyber physical cognitive unknown Etc so if we're doing brick collage on the edges now we have the glue or the sticky tape or whatever it is and then last point was on the peacocks and pyramids it's not so simple as just an Adaptive buffer even throwing a pleiotropy so the consequences on different phenotypes uh aside because it becomes integrated as the kind of thing I am and so yes it's like the Eraser is the capacity of the pencil but we don't rub our erasers all the way down we're not always on our last life in the video game and a symbol that on one hand the energy could have been diverted to some other purpose that symbol can actually become an introduction at even the mythopoetic and therefore actually lose its capacity to be an Adaptive buffer at one level and become a pivotal player at another level the buffer I think that is that's also like related to that Instagram problem right where it's like and it's particularly with respect to the abstraction of that buffer as well right so like if you abstract the buffer as wealth the buffer is inferior signal of underlying quality of behavior to accumulating resources uh over and above the requisite level of short-term consumption or metabolic need right that's the evolutionary Instinct we have for sort of appreciating someone who has quote unquote gathered well at least by means that were constructive or generative um but then you can you know to the extent that becomes associated with identity right and then you look at the signals as that wealth manifests in a particular culture as the proxy measurement for the underlying for the underlying phenomenon then you can start to get into the manipulation of those signals if someone understands and observes others behaving as if the signals were the process of production of that capacity for well that the capacity production of generating wealth or useful byproducts then you can see

again this immersion parasitism uh element dropping in there because people will try to middleman that process and say if if I can get you to treat me as if I am that person maybe I can get access to whatever Associated uh privileges come alongside that so so yeah 100 that's a um that identity Association and and all of the games that it opens up are is essentially Central what's your main to what we're talking about what we were awesome and uh a closing thought the discussion made me think of a July 4th fireworks for information what does it look like when the Weaponry is turned into livingry and when there can be a symbolic ritual around the utilization of that dual functional technology within the boundaries across those levels yeah I mean we're in a beautiful time for that right now I think so much of the emerging technology the generative Technologies out there especially under the auspices of AI or ml the capacity to use that generative toolkit and it's its ability to allow people to uh a much broader swath of people to produce potentially resonant artistic and cultural artifacts that are Pointers to something they find unique or interesting or insightful that might then spread if it aligns with a given moment or need and that's particularly fascinating to me as a as a catalyst for increasing our capacity to think collectively uh you know transcending just the verbal domain and bringing in other sensory modalities uh aligned with the word I mean that's the funny thing about these llms right like they're large language models and you know language can act as that fabric as that as that you know language as the net neural network itself as the as the computational network between these other sensory apparatus and and the ability to to produce artifacts that bring people into forms of alignment or phenomenological resonance that just would not have otherwise existed um now you know people are there are potential downsides this as well just like anything but you know I'm very excited about that because I look at you know the all of the art that I've seen uh this new generative artwork it is like you know very much informational fireworks right people are able to take five ideas they heard today you could go you know based on this podcast you could basically say like take any subsection of that produce a series of generative artworks and then associate that lineage or like that like series of generative art with this as a network a visual Network that could be a dynamic placeholder you know that could attract attention uh automatically from the potential Network that resonates with that artwork right and and and tap into the latent space of this conversation that none of us hold explicitly in our heads but we're all pointing to it based on our words and our conversations and the kind of things we're trying to talk about here like there is a latent space representation uh possible so yeah I mean I think we're in fascinating times for that and using that as a tool constructively and wisely and I'm looking forward to

seeing seeing that part of this unfold thank you Matthew thank you to Jordan till next time
yeah I think so always a pleasure